



Montag 06.07.2026
Tag 11

Von KI erstellt, von Menschen kontrolliert.

Guten Morgen!

GitHub Repo <https://github.com/www42/a5>



Azure in 5 Wochen

Woche 1	(22. – 26. Juni)	Trainer: Thomas Jäkel	Online	Intro, Identity, Governance
Woche 2	(29. – 03. Juli)	Trainer: Thomas Jäkel	HD	Tools, Networking
Woche 3	(06. – 10. Juli)	Trainer: Thomas Jäkel	HD/MA	Storage, Compute 1
→ Woche 4	(13. – 17. Juli)	Trainer: Jörg Heinlein	Online	Compute 2, Monitoring, Backup
Woche 5	(20. – 24. Juli)	Trainer: Thomas Jäkel	HD/MA	Prüfungsvorbereitung, Prüfung

AZ-104

Learning Paths:

- Manage identities and governance in Azure
- Prerequisites for Azure administrators – CloudShell and templates
- Configure and manage virtual networks for Azure administrators
- Implement and manage storage in Azure
- Deploy and manage Azure compute resources
- Monitor and back up Azure resources

Heute:

- Zusammenfassung Azure Networking
- Lab 05 (AZ-104) Skillable

-
- Storage Accounts
 - Lab 07 (AZ-104) Skillable

Azure for Students
88€ / 100\$
1 Jahr

Review Azure Networking

<https://learn.microsoft.com/en-us/azure/networking/>



- Virtual Network / Subnet
- Peering
- Hub & Spoke Topology
- Private Endpoint
- Public IP Address
- Standard Routing / UDR
- Service Endpoint
- Virtual Network Gateway
- NSG
- ASG
- Load Balancer
- Application Gateway
- Azure Virtual WAN
- Express Route
- DDos Protection ✓
- Network Watcher
- Private DNS Zone

Daniel ✓
Samuel ✓
Eugen ✓
Leena

Deniz ✓
Kati ✓
Alex ✓
Ferhat ✓
Wahid ✓
Marti ✓
Sercan ✓

Omar ✓

Matthias ✓
Semih ✓

(Richard Feynman)

Docs / Gemini / Google

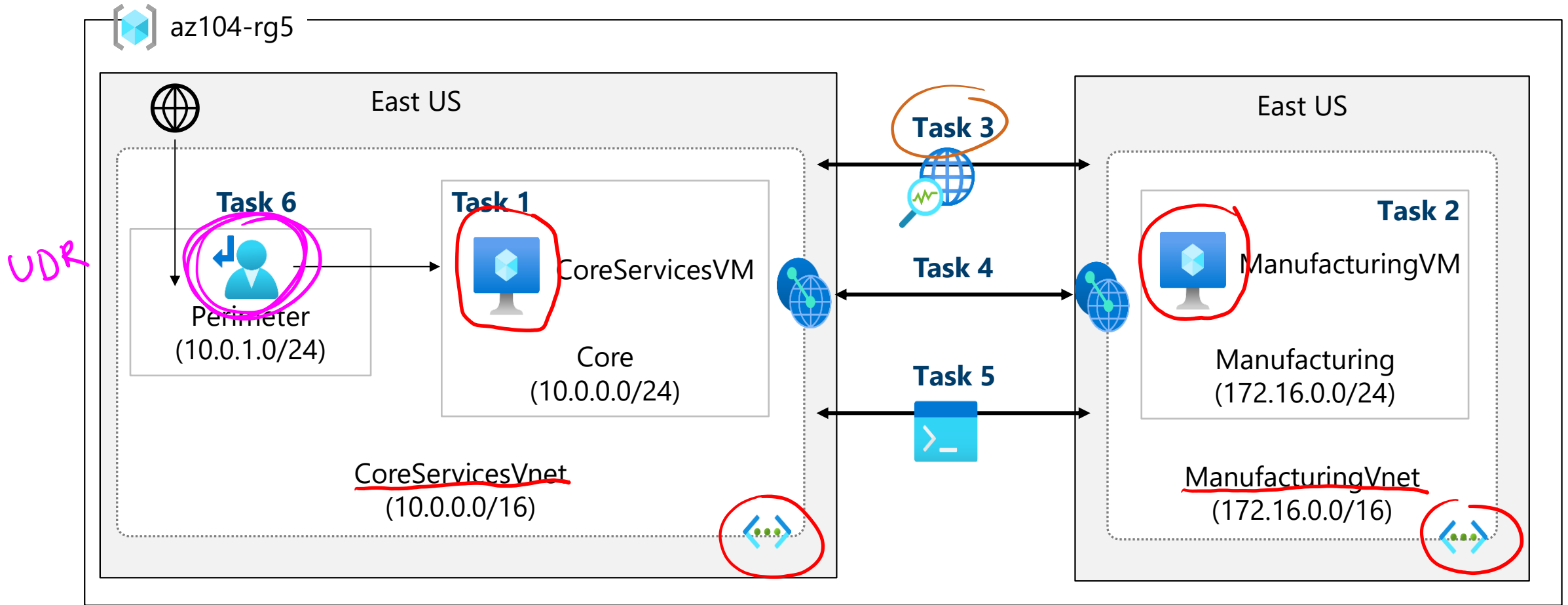
g35 Uhr

Lab 05 - Implement Intersite Connectivity



Lab 05 – Architecture diagram

Test-NetConnection das bessere ping

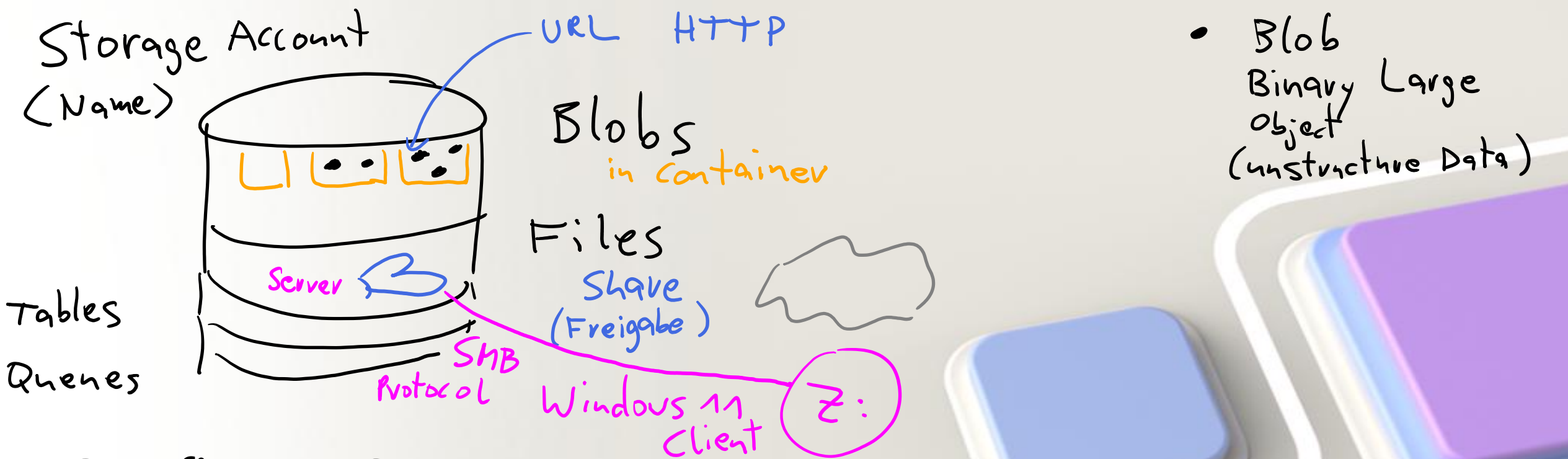


Task 3: No Peering → No Ping
Task 4: Peering! Ping!

AZ-104

 Implement and manage storage in Azure





Configure Storage Accounts

Server Message Blocks (SMB)

SMB

Network File System (Linux)

NFS

Explore Azure Storage Services

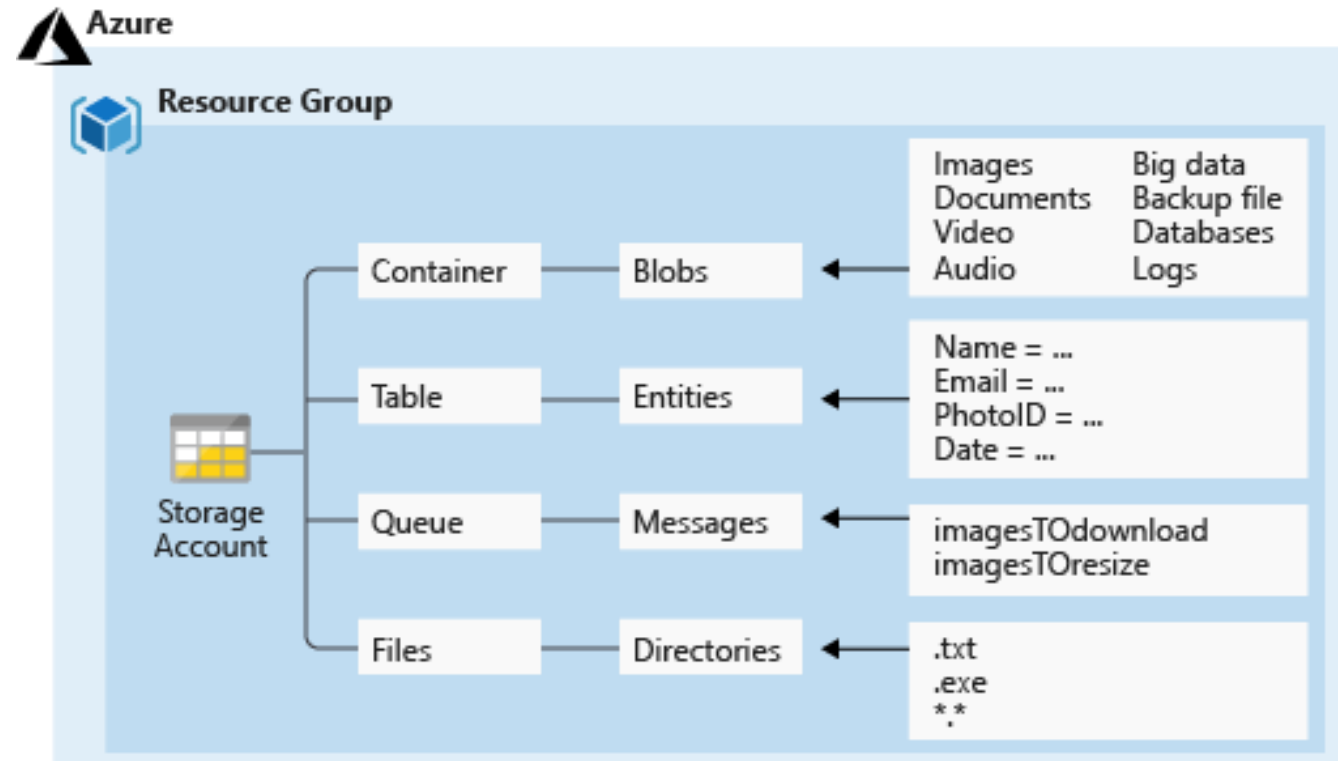
A service that you can use to store files, messages, tables, and other types of information

Azure Containers: A massively scalable object store for text and binary data

Azure Tables: Ideal for storing structured, non-relational data

Azure Queues: A messaging store for reliable messaging between application components

Azure Files: Managed file shares for cloud or on-premises deployments

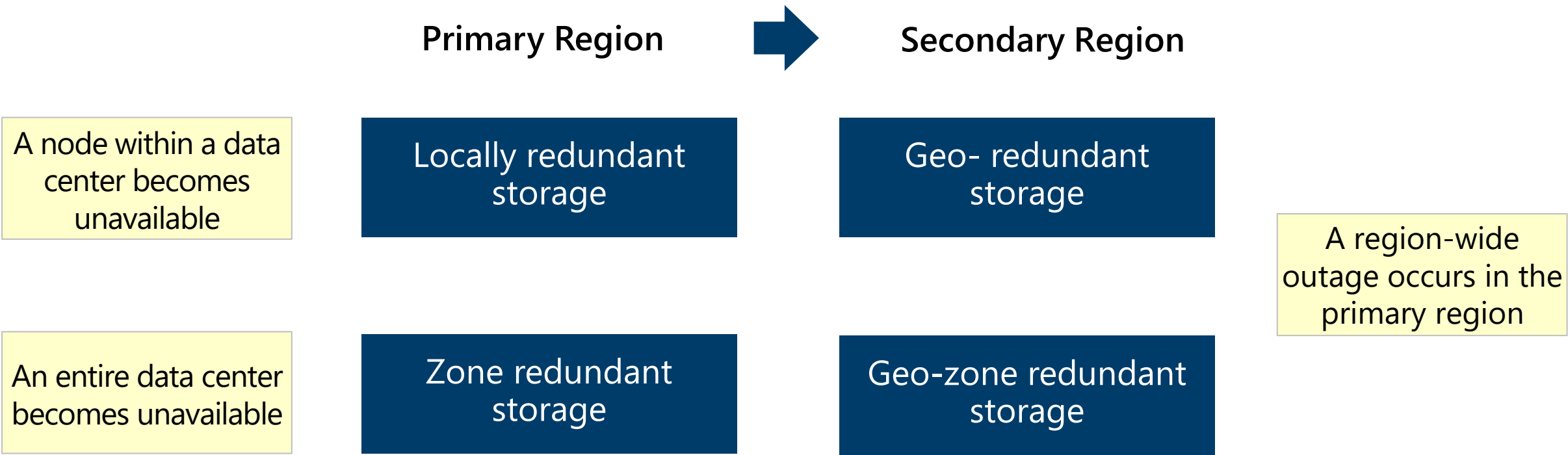


Determine Storage Account Kinds

All storage accounts are encrypted using Storage Service Encryption (SSE) for data at rest

Storage Account	Recommended usage
Standard general-purpose v2	Most scenarios including Blob, File, Queue, Table, and Data Lake Storage.
Premium block blobs	Block blob scenarios with high transactions rates, or scenarios that use smaller objects or require consistently low storage latency.
Premium file shares	Enterprise or high-performance file share applications.
Premium page blobs	Premium high-performance page blob scenarios.

Determine Replication Strategies



Access Storage

Every object has a unique URL address – based on account name and storage type

Container service: `https://mystorageaccount.blob.core.windows.net`

Table service: `https://mystorageaccount.table.core.windows.net`

Queue service: `https://mystorageaccount.queue.core.windows.net`

File service: `https://mystorageaccount.file.core.windows.net`

- If you prefer you can configure a custom domain name

CNAME record	Target
blobs.contoso.com	contosoblobs.blob.core.windows.net

Secure Storage Endpoints

Firewalls and Virtual Networks restrict access to the Storage Account from specific Subnets on Virtual Networks or public IP's

Subnets and Virtual Networks must exist in the same Azure Region or Region Pair as the Storage Account

Firewalls and virtual networks

Custom domain

 Save  Discard  Refresh

Public network access

- ☒ Enabled from all networks
- ☐ Enabled from selected virtual networks and IP addresses
- ☐ Disabled

 All networks, including the internet, can access this storage account. [Learn more](#) 

Network Routing

Determine how you would like to route your traffic as it travels from its source to an Azure endpoint. Microsoft routing is recommended for most customers.

Routing preference ⓘ

- ☒ Microsoft network routing ☐ Internet routing

 The current combination of storage account kind, performance, replication, and location does not support network routing.

Configure Blob Storage

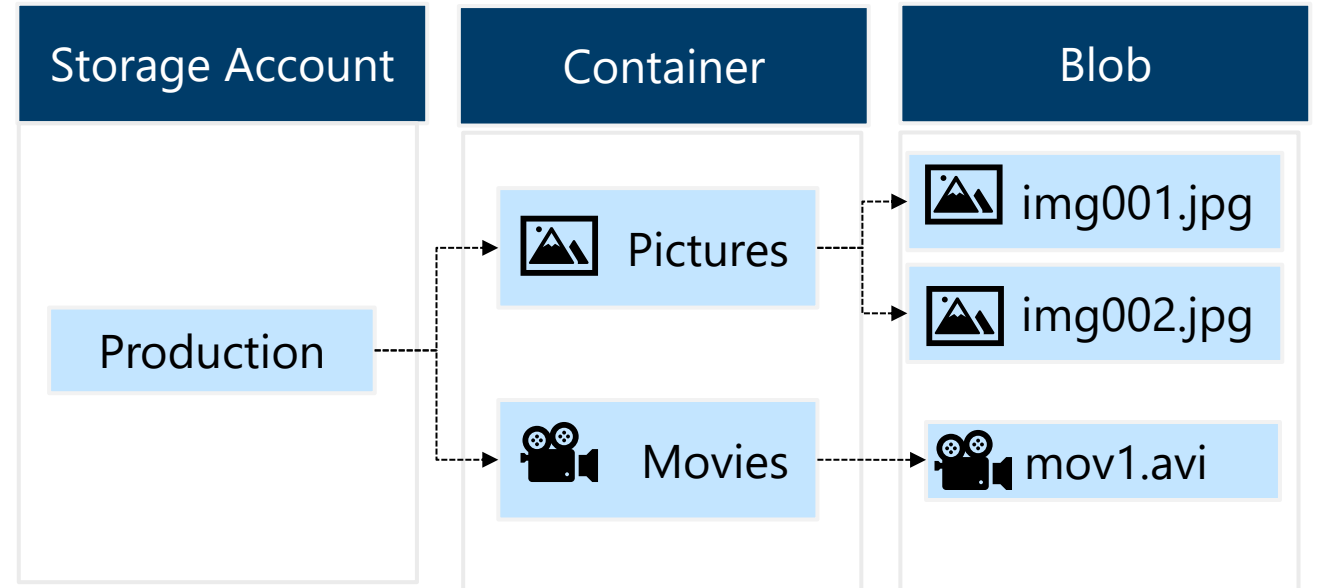


Implement Blob Storage

- Stores unstructured data in the cloud
- Can store any type of text or binary data
- Also referred to as *object storage*

Common uses:

- Serving images or documents directly to a browser
- Storing files for distributed access
- Streaming video and audio
- Storing data for backup and restore, disaster recovery, archiving
- Storing data for analysis by an on-premises or Azure-hosted service



Create Blob Containers

- All blobs must be in a container
- Accounts have unlimited containers
- Containers can have unlimited blobs
- Restrict access using the public access level

[+ Container](#) [🔒 Change access level](#) [🔄 Refresh](#) | [🗑️ Delete](#)

New container

Name *

container01 ✓

Public access level ⓘ

Private (no anonymous access) ^

Private (no anonymous access)

Blob (anonymous read access for blobs only)

Container (anonymous read access for containers and blobs)

OK

Cancel

Create Blob Access Tiers

Hot tier – Data that is accessed or modified frequently

Cool tier – Data that is infrequently accessed or modified and stored for at least 30 days

Cold tier – Data that is infrequently accessed or modified and stored for at least 90 days

Archive – Data that can tolerate several hours of retrieval latency and will remain in the Archive tier for at least 180 days

Change tier

infoicon.jpg



Optimize storage costs by placing your data in the appropriate access tier. [Learn more](#)

Access tier

Hot (Inferred)



Hot (Inferred)

Cool

Cold

Archive

Add Blob Lifecycle Management Rules

- Transition blobs to a cooler storage tier to optimize for performance and cost
- Delete blobs at the end of their lifecycle
- Apply rules for lifecycle management tasks

Add a rule ...

✓ Details 2 Base blobs 3 Filter set

Lifecycle management uses your rules to automatically move blobs to cooler tiers or to delete them. If you create multiple rules, the associated actions must be implemented in tier order (from hot to cool storage, then archive, then deletion).

If

Base blobs were *

☒ Last modified

☐ Created

More than (days ago) *

↓

Then

Delete the blob

Move to cool storage
For infrequently accessed data that you want to keep on cool storage for at least 30 days.

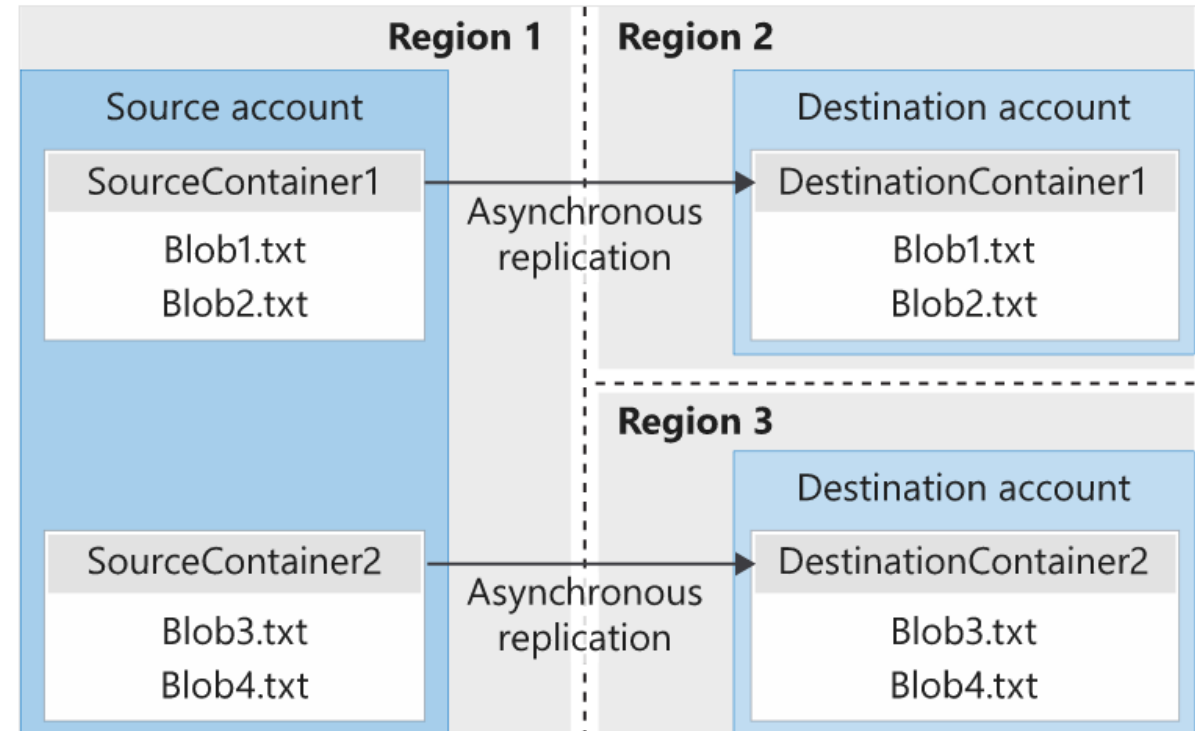
Move to cold storage
For rarely accessed data that you want to keep for at least 90 days.

Move to archive storage
Use if you don't need online access and want to keep the object for 180 days or longer.

Delete the blob
Deletes the object per the specified conditions.

Determine Blob Object Replication

- Asynchronous to any other Region
- Minimizes latency for read requests
- Increases efficiency for compute workloads
- Optimizes data distribution
- Optimizes costs



Configure Azure Files



Compare Storage for File Shares and Blob Data

Feature	Description	When to use
Azure Files	SMB interface, client libraries, and a REST interface that allows access from anywhere to stored files	• Lift and shift an application to the cloud • Store shared data across multiple virtual machines • Store development and debugging tools that need to be accessed from many virtual machines
Azure Blobs	Client libraries and a REST interface that allows unstructured data (flat namespace) to be stored and accessed at a massive scale in block blobs	• Support streaming and random-access scenarios • Access application data from anywhere

Manage File Shares

- Windows – ensure port 445 is open
- Linux – mount the drive
- MacOS – mount the drive
- Secure transfer required – SMB 3.0 encryption

Connect

publicwebsite

Windows

Linux

macOS

To connect to this Azure file share from Windows, choose from the following authentication methods and run the PowerShell commands from a normal (not elevated) PowerShell terminal:

Drive letter

Z

Authentication method

☐

Active Directory or Microsoft Entra

☒

Storage account key



Connecting to a share using the storage account key is only appropriate for admin access. Mounting the Azure file share with the Active Directory or Microsoft Entra identity of the user is preferred. [Learn more](#)

Show Script

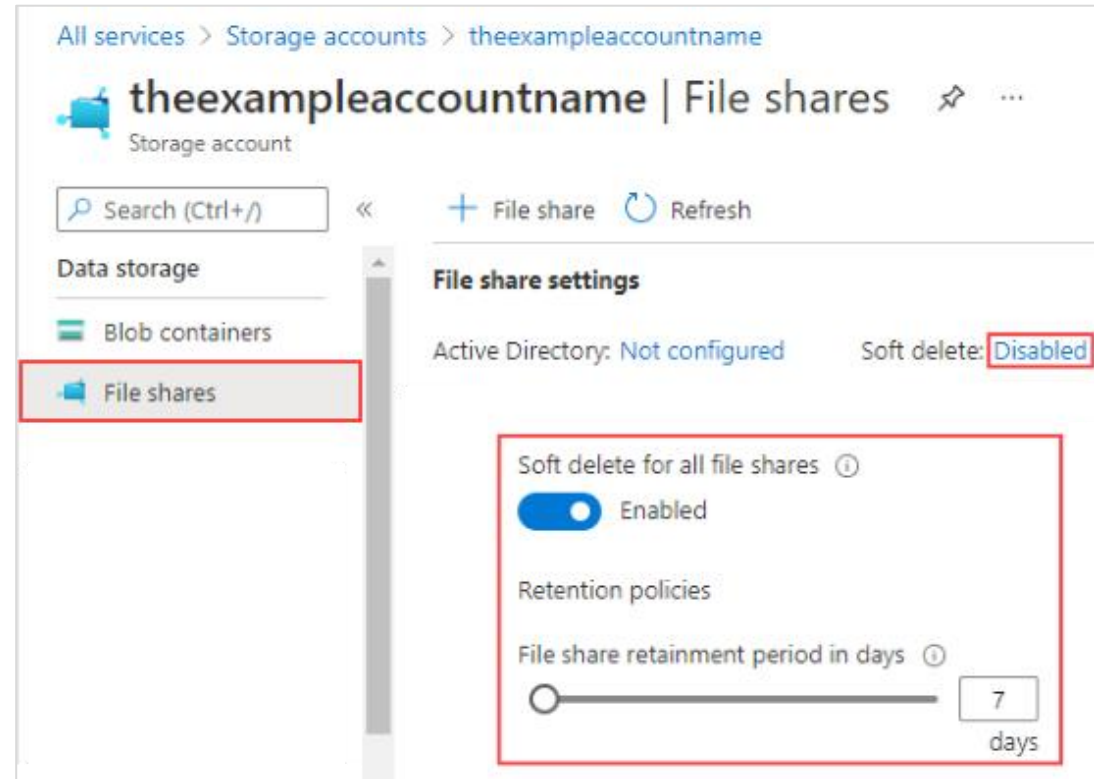
Create File Share Snapshots

- Protection against application error and data corruption
- Protection against accidental deletions or unintended changes
- Support backup and recovery
- Incremental snapshot that captures the share state at a point in time
- Snapshot at the *file share level*, and restore at the *file level*
- Is read-only copy of your data

+ Add snapshot ↻ Refresh 🗑 Delete		
Name	Date created	Initiator
☐ 2020-03-12T00:58:38.0000000Z	3/11/2020, 8:58:38 PM	-

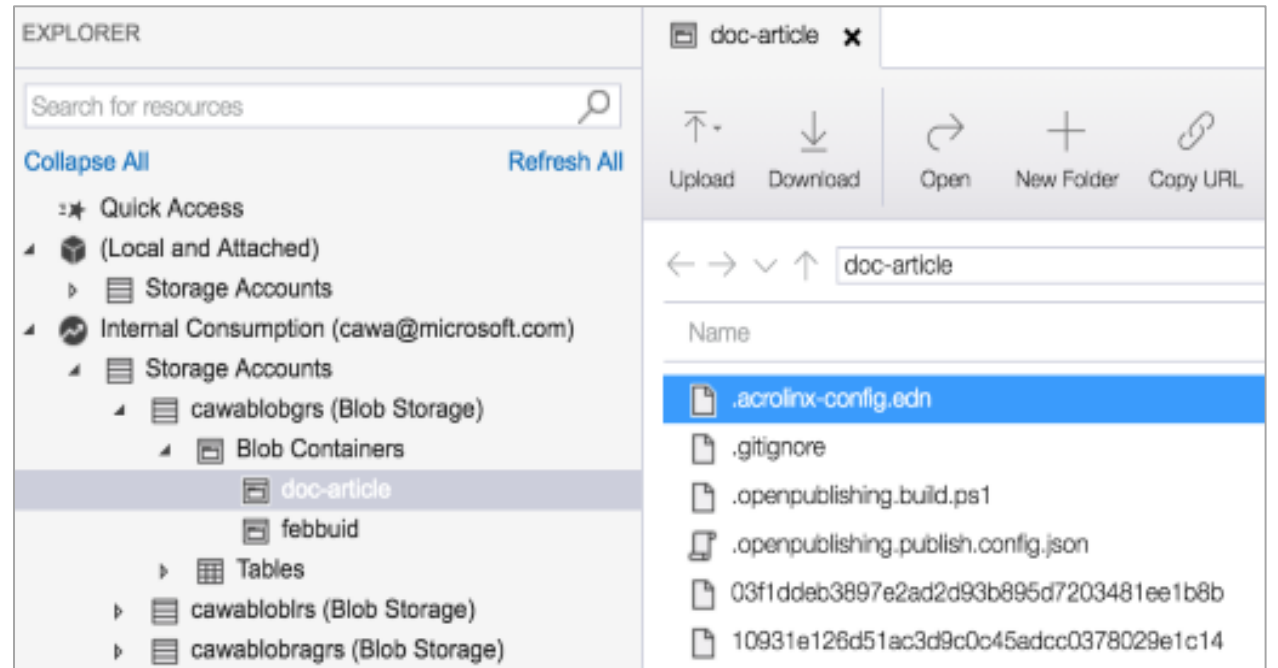
Implement Soft Delete for Azure Files

- Recovery from accidental data loss
 - Major change or upgrade scenarios
 - Business continuity – ransomware situations
 - Data compliance retention
-
- Enabled at the storage account level
 - Transitions content to a soft deleted state
 - Provides a retention period of 1 and 365 days
 - Works on new or existing file shares
 - Doesn't work for NFS shares



Use Azure Storage Explorer

- Download and install
- Access multiple accounts and subscriptions
- Create, delete, view, edit storage resources
- View and edit Blob, Queue, Table, File, Cosmos DB storage and Data Lake Storage
- Obtain shared access signature (SAS) keys
- Available for Windows, Mac, and Linux



Also consider portal-based Azure Storage Browser and Azure Storage Mover

Configure Storage Security



Review Storage Security Strategies



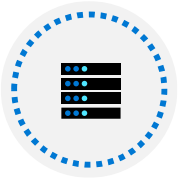
Storage Service Encryption



Authentication with Entra ID and RBAC



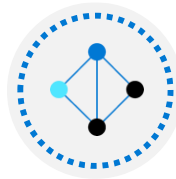
Client-side encryption, HTTPS, and SMB 3.0 for data in transit



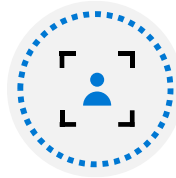
Azure disk encryption



SAS "Mit Freunden teilen"
Shared Access Signatures – delegated access



Shared Key – encrypted signature string



Anonymous access to containers and blobs

Create Shared Access Signatures

- Provides delegated access to resources
- Grants access to clients without sharing your storage account keys
- The account SAS delegates access to resources in one or more of the storage services
- The service SAS delegates access to a resource in just one of the storage services

The screenshot shows the 'Create Shared Access Signature' form in the Azure portal. Handwritten blue annotations include a cylinder labeled 'Storage Account' and the text 'key 1' and 'key 2'. A blue oval highlights the 'Signing key' dropdown menu, which is currently set to 'Key 1'. A red oval highlights the 'Permissions' dropdown menu, which is currently set to 'Read'. The 'Start and expiry date/time' section shows a start date of '02/01/2021' and an expiry date of '02/02/2021', both with red underlines. The 'Allowed IP addresses' field contains the text 'for example, 168.1.5.65 or 168.1.5.65-168.1....'. The 'Allowed protocols' section has 'HTTPS' selected. A blue button at the bottom is labeled 'Generate SAS token and URL'.

Signing method ⓘ
☒ Account key ☐ User delegation key

Signing key ⓘ
Key 1 ▼

Permissions ⓘ
Read ▼

Start and expiry date/time ⓘ
Start
02/01/2021
(UTC-08:00) Coordinated Universal Time-08 ▼
Expiry
02/02/2021
(UTC-08:00) Coordinated Universal Time-08 ▼

Allowed IP addresses ⓘ
for example, 168.1.5.65 or 168.1.5.65-168.1....

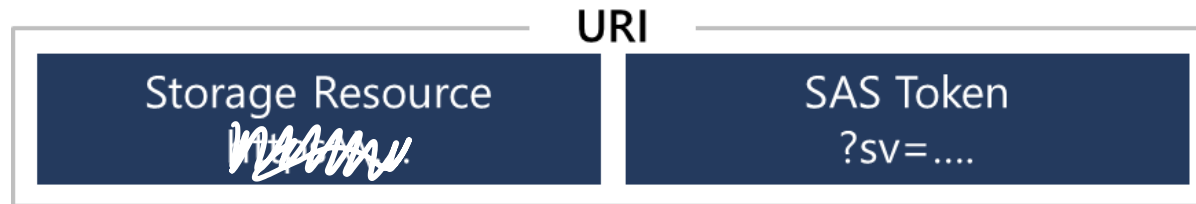
Allowed protocols ⓘ
☒ HTTPS ☐ HTTP

Generate SAS token and URL

key 1
key 2
Storage Account

Identify URI and SAS Parameters

- A SAS is a signed URI that points to one or more storage resources
- Consists of a storage resource URI and the SAS token



<https://myaccount.blob.core.windows.net/?sp=r&st=2020-05-11T18:31:43Z&se=2020-05-12T02:31:43Z&spr=https&sv=2019-10-10&sr=b&sig=j0qABJZHfUVEBQ3yVn7kWiCKl00sxCiK1rzEchfAz8U%3D>

Includes parameters for resource URI, storage services version, services, resource types, start time, expiry time, resource, permissions, IP range, protocol, signature

TO THE LAB ZONE →

NO THROUGH ROUTE

Berlin
Europa
Dublin

SPEEDY ROUTE

4²

lounge

Microsoft
tech-ed
2010

lounge



software diagnostics

E94

SD Studio

for Team Foundation Server
Visualize your code in
a software map.

SD Developer Edit

for C and C++ for Visual
Trace dynamically your code
Visualize the control flow.

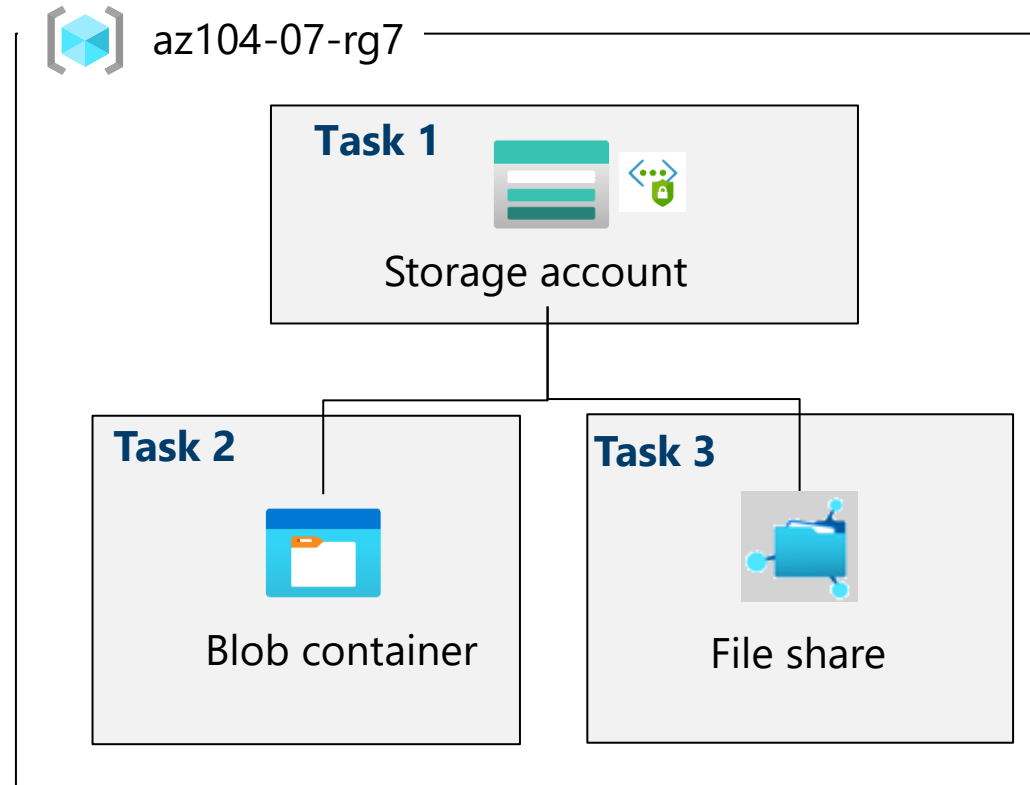
software
diagnostics

DataCore
SOFTWARE

Lab 07 – Manage Azure Storage



Lab 07 – Architecture diagram



End of presentation



